YOUNES MOUHIB

Mathematician: optimization, data analysis, rigorous evaluation

Affoltern am Albis (Zurich), Switzerland · +41 78 212 46 74 · younes.mouhib@hotmail.fr

LinkedIn · GitHub (github.com/ysmouhib) · Website (ysmouhib.github.io)

MSc mathematician (ETH Zurich) who models hard problems and measures honestly: neural networks combined with constraint solvers, risk models tested on unseen data, and LLM agents gated by formal verification. Looking for a first industry role where mathematical depth and evaluation discipline create direct value.

Education

MSc in Mathematics, ETH Zurich **08.2026**

Network and Integer Optimization, Graph Theory, Structural Graph Theory, Algebraic Methods in Combinatorics

BSc in Mathematics, EPFL **07.2023**

Probability, Statistics, Stochastic Processes, Numerical Analysis, Functional Analysis, Discrete Optimization

Applied Projects

Football injury risk (Python, scikit-learn) **2026**

- Weekly injury-risk scores for all players in Europe's top five leagues, built from public data only. Calibrated logistic regression, tested once on the fully unseen 2024/25 season: top-10% risk group injured 6× as often as the bottom 10% (19.5% vs 3.1%).

Sudoku vision (PyTorch, ONNX, SAT/MaxSAT) **2026**

- A CNN reads handwritten grids, a SAT solver solves them, MaxSAT repairs misread digits. Fully correct grids improved from 75% to 99% on 1,000 held-out puzzles; ablation across three classifiers isolates each component's contribution.

NEAR momentum bot (Python) **2026**

- Cost-aware backtesting with walk-forward validation for a trading strategy. Documents that the v1 edge did not survive 0.24% round-trip costs (82% out-of-sample loss) and rebuilds the evaluation so small samples are read honestly - a study in evaluation discipline.

Brigade, multi-agent LLM prover (Python, FastAPI, Lean 4) **2024 - 2026**

- LLM agents plan, write and repair proofs; only an independent Lean 4 + Mathlib checker can accept them. Dockerized, continuous integration, in-browser demo.

Common Ground (Python, Lean 4, React) **2026**

- First-order logic engine certifying whether conclusions follow from premises, with Lean-checked proofs. Matches FOLIO labels on 195/204 validation examples (95.6%).

Research

Master thesis, ETH Zurich (Prof. Dr. Lorenz Halbeisen) **07.2026**

A symmetry reduction turned a brute-force search over 10 billion configurations into a 253-variable SAT model solved in seconds, setting new record lower bounds for the Hales–Jewett numbers (HJ(3,3): 14 to 22, HJ(4,2): 12 to 14). Every bound re-verified in CI by standalone checkers.

Preprint (sole author) **06.2026**

arXiv:2606.22155 - Lower Bounds for the Hales–Jewett Numbers via Symmetric and One-Weight Colorings

Seminar talk, ETH Zurich (Prof. Dr. Benny Sudakov); Bachelor thesis, EPFL **2025 / 2023**

Graph Regularity Method (two-hour talk); Subadditive Ergodic Theorem and its applications

Experience

Course instructor, Topology (PVK), ETH Zurich **06.2026**

Designed and taught a four-day exam-preparation course for 20+ students

Homework tutor, mathematics and science, Lausanne **2021 - 2023**

Individual study plans for pupils in grades 9–11; studies self-financed through tutoring and part-time work

Skills

Languages Python, SQL, MATLAB, Lean 4

ML & data PyTorch, scikit-learn, pandas, NumPy

Optimization Gurobi, SAT/MaxSAT (PySAT, CryptoMiniSat), constraint modeling

Tooling Git, Docker, CI (GitHub Actions), Linux, LaTeX

Languages French (native), Moroccan Arabic (native), English (C1), Spanish (B2), German (A1, improving)

Interests: chess, martial arts and classical guitar (10+ years each). References available upon request.